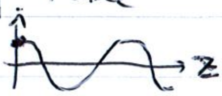
Solutions to MP204 Problem Set 6

(1)

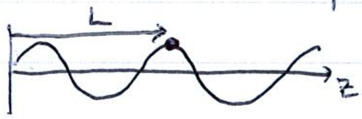
P.1

This problem covers the basics of propagating EM waves: direction of travel, directions of both the electric and magnetic field, etc.

(a) The direction is determined by the time dependence: any travelling wave depends on the quantity $\hat{n}\cdot\vec{r} - ct$, where $\hat{n}$ determines the direction of travel. In this case, where we see $z - ct$ appearing, this means it's travelling $\boxed{\text{in the positive } z \text{ direction}}$. To see this, sketch $|\vec{E}|$ at $t = 0$: it looks like this:



Now, let time run forward a bit, say to $t = L/c$, where L is some length. The point that used to be at $z = 0$ is now at the point $z - c(L/c) = z - L = 0$, i.e. at $z = L$.



This means that the shape has travelled to the right a distance L, so it is indeed moving in the positive z-direction.

(b) Recall that the direction of motion, the electric field and the magnetic field point in the same orientation as a right-handed coordinate system. We just determined that the direction of motion of this wave is in the $\hat{k}$ direction, and $\vec{E} = 3E_0\cos(z - ct)\,\hat{\imath}$, so this determines the direction of $\vec{B}$: the cross product of the first two vectors in a right-handed coordinate gives the direction of the third, so $\vec{B}$ must point in the $\hat{k}$ (propagation direction) $\times\, \hat{\imath}$ (electric field direction) $= \hat{\jmath}$. Thus, $\boxed{\vec{B} \text{ points in the } y\text{-direction.}}$ Thus, the constant vector $\vec{B}_0$ is $|\vec{B}_0|\,\hat{\jmath}$. Now, we should recall that a travelling wave has a magnetic field whose magnitude is that of the electric field divided by c: the amplitude of the $\vec{E}$-field is $3E_0$, so $\boxed{|\vec{B}| = 3E_0/c}$. Thus,

$\vec{B} = \dfrac{3E_0}{c}\cos(z - ct)\,\hat{\jmath}$ is the magnetic field.

We can see this another way as well: $\vec{\nabla}\times\vec{E} = -\dfrac{\partial \vec{B}}{\partial t}$ must be satisfied. Using our $\vec{E}$, we have

$\vec{\nabla}\times\vec{E} = \dfrac{\partial E_x}{\partial z}\,\hat{\jmath}$ (because $\vec{E}$ is only in the x-direction and only depends on z) $= -3E_0\sin(z - ct)\,\hat{\jmath}$. But

$\dfrac{\partial \vec{B}}{\partial t} = c\vec{B}_0\sin(z - ct)$. $\vec{\nabla}\times\vec{E} = -\dfrac{\partial \vec{B}}{\partial t}$ thus gives

$-3E_0 \sin(z-ct)\,\hat{j} = -c\vec{B}_0 \sin(z-ct)$, or $\vec{B}_0 = \frac{3E_0}{c}\hat{j}$, as we saw above. So there's more than one way to find the magnetic field from the electric field.

P. 2

(a) The wave eq'n is, of course, $\frac{1}{c^2}\frac{\partial^2\vec{E}}{\partial t^2} = \nabla^2\vec{E}$. Technically, this is three eq'ns, one for each of the x, y and z components of $\vec{E}$. However, the $\vec{E}$ given has only a z-component, so we need only check that $\frac{1}{c^2}\frac{\partial^2 E_z}{\partial t^2} = \nabla^2 E_z$. Let's do that:

$$\frac{1}{c^2}\frac{\partial^2 E_z}{\partial t^2} = \frac{1}{c^2}\frac{\partial}{\partial t}\left[\frac{\partial}{\partial t} E_0\cos(ky)\cos(\omega t)\right]$$
$$= -\frac{1}{c^2}\frac{\partial}{\partial t}\left[E_0\omega\cos(ky)\sin(\omega t)\right]$$
$$= -\frac{1}{c^2}\left[E_0\omega^2\cos(ky)\cos(\omega t)\right]$$
$$= -\frac{E_0\omega^2}{c^2}\cos(ky)\cos(\omega t)$$

$$\nabla^2 E_z = \frac{\partial^2 E_z}{\partial x^2} + \frac{\partial^2 E_z}{\partial y^2} + \frac{\partial^2 E_z}{\partial z^2}$$
$$= \frac{\partial}{\partial y}\left[\frac{\partial}{\partial y} E_0\cos(ky)\cos(\omega t)\right]$$
$$= \frac{\partial}{\partial y}\left[-kE_0\sin(ky)\cos(\omega t)\right]$$
$$= -k^2E_0\cos(ky)\cos(\omega t).$$

But remember that $\lambda\nu = c$: this means that $k = \frac{2\pi}{\lambda} = \frac{2\pi\nu}{c} = \frac{\omega}{c}$, so $-k^2E_0\cos(ky)\cos(\omega t) = -\frac{\omega^2}{c^2}E_0\cos(ky)\cos(\omega t)$, which is $\frac{1}{c^2}\frac{\partial^2 E_z}{\partial t^2}$. Thus, $\boxed{\frac{1}{c^2}\frac{\partial^2\vec{E}}{\partial t^2} = \nabla^2\vec{E}}$ and $\vec{E}$ satisfies the wave eq'n.

(b) If we know $\vec{E}$, we can compute $\vec{\nabla}\times\vec{E}$. This is supposed to be $-\frac{\partial\vec{B}}{\partial t}$, so if we integrate $-\vec{\nabla}\times\vec{E}$ with respect to time, we get $\vec{B}$. Let's do it:

$$-\vec{\nabla}\times\vec{E} = -\frac{\partial E_z}{\partial y}\hat{i}$$ (since $\vec{E}$ is only in the z-direction and only depends on y)

$$= -\frac{\partial}{\partial y}\left[E_0\cos(ky)\cos(\omega t)\right]\hat{i}$$
$$= E_0 k\sin(ky)\cos(\omega t)\,\hat{i}.$$

Integrating this with respect to t gives $\vec{B}$:

$$\vec{B} = \int E_0 k\sin(ky)\cos(\omega t)\,dt\,\hat{i} = \frac{E_0 k}{\omega}\sin(kx)\sin(\omega t)\,\hat{i}$$
$$= \boxed{\frac{E_0}{c}\sin(ky)\sin(\omega t)\,\hat{i}}$$

where I have again used $k = \omega/c$.

(c) The trig identity for this problem will be

$$\cos\alpha\cos\beta = \frac{1}{2}\left[\cos(\alpha+\beta) + \cos(\alpha-\beta)\right]$$

because it means we can rewrite $\vec{E}$ as

$$\vec{E} = E_0 \left\{ \frac{1}{2}\cos(ky+\omega t) + \frac{1}{2}\cos(ky-\omega t) \right\} \hat{k}$$

$$= \left\{ \frac{E_0}{2}\cos(k(y+ct)) + \frac{E_0}{2}\cos(k(y-ct)) \right\} \hat{k}$$

because, once again, $\omega = kc$. Thus, we see it consists of one part that depends only on the combination $y+ct$ and another that depends on $y-ct$. In other words, if

$$\boxed{f_+(y) = f_-(y) = \frac{E_0}{2}\cos(ky)}$$

then $\vec{E} = \left[f_+(y+ct) + f_-(y-ct)\right]\hat{k}$.

P.3

The energy density is $u = \frac{1}{2}\epsilon_0|\vec{E}|^2 + \frac{1}{2\mu_0}|\vec{B}|^2$ and the Poynting vector is $\vec{S} = \frac{1}{\mu_0}\vec{E}\times\vec{B}$, and since we know both $\vec{E}$ and $\vec{B}$ from Problem 1 and 2, we can find both of these.

(a) The energy density is

$$u = \frac{1}{2}\epsilon_0\left|3E_0\cos(z-ct)\hat{\imath}\right|^2 + \frac{1}{2\mu_0}\left|\frac{3E_0}{c}\cos(z-ct)\hat{\jmath}\right|^2$$

$$= \frac{1}{2}\epsilon_0\left(9E_0^2\cos^2(z-ct)\right) + \frac{\epsilon_0 c^2}{2}\left(\frac{9E_0^2}{c^2}\cos^2(z-ct)\right)$$

$$= \boxed{9\epsilon_0 E_0^2\cos^2(z-ct)}$$

where we used $\mu_0 = 1/\epsilon_0 c^2$ to put it into the simplest form possible. (You don't have to do this if you don't want to.)

The Poynting vector is

$$\vec{S} = \frac{1}{\mu_0}\vec{E}\times\vec{B} = \frac{1}{\mu_0}\left(3E_0\cos(z-ct)\hat{\imath}\right)\times\left(\frac{3E_0}{c}\cos(z-ct)\hat{\jmath}\right)$$

$$= \boxed{\frac{9E_0^2}{\mu_0 c}\cos^2(z-ct)\hat{k} = 9c\epsilon_0 E_0^2\cos^2(z-ct)\hat{k}.}$$

(Once again, note that $\vec{S}$ points in the direction of propagation for a travelling wave.)

(b) For the fields in P.2, we get

$$u = \frac{1}{2}\epsilon_0|\vec{E}|^2 + \frac{1}{2\mu_0}|\vec{B}|^2$$

$$= \frac{1}{2}\epsilon_0\left|E_0\cos(ky)\cos(\omega t)\hat{k}\right|^2 + \frac{\epsilon_0 c^2}{2}\left|\frac{E_0}{c}\sin(ky)\sin(\omega t)\hat{i}\right|^2$$

$$= \boxed{\frac{1}{2}\epsilon_0 E_0^2\left[\cos^2(ky)\cos^2(\omega t) + \sin^2(ky)\sin^2(\omega t)\right]}$$

for the energy density, and

$$\vec{S} = \frac{1}{\mu_0}\vec{E}\times\vec{B} = \epsilon_0 c^2\left(E_0\cos(ky)\cos(\omega t)\hat{k}\right)\times\left(\frac{E_0}{c}\sin(ky)\sin(\omega t)\hat{i}\right)$$

$$= \boxed{c\epsilon_0 E_0^2\sin(ky)\cos(ky)\sin(\omega t)\cos(\omega t)\hat{j}}$$

as the Poynting vector.

P.4

We want to get an idea of the eq'ns which govern electromagnetism inside a conductor. If the conductor is electrically neutral, then $\rho = 0$, but $\vec{J}$ is not: instead, we have Ohm's Law $\vec{J} = \sigma\vec{E}$. If we write Maxwell's eq'ns for this situation, we get

$$\vec{\nabla}\cdot\vec{E} = 0,\quad \vec{\nabla}\cdot\vec{B} = 0,\quad \vec{\nabla}\times\vec{E} = -\frac{\partial\vec{B}}{\partial t},\quad \vec{\nabla}\times\vec{B} = \mu_0\sigma\vec{E} + \mu_0\epsilon_0\frac{\partial\vec{E}}{\partial t}$$

Let's take the curl of the third of these:

$$\vec{\nabla}\times(\vec{\nabla}\times\vec{E}) = \vec{\nabla}\times\left(-\frac{\partial\vec{B}}{\partial t}\right) \Rightarrow \vec{\nabla}(\vec{\nabla}\cdot\vec{E}) - \nabla^2\vec{E} = -\frac{\partial}{\partial t}(\vec{\nabla}\times\vec{B})$$

Using $\vec{\nabla}\cdot\vec{E} = 0$ and $\vec{\nabla}\times\vec{B} = \mu_0\sigma\vec{E} + \mu_0\epsilon_0\frac{\partial\vec{E}}{\partial t}$ gives

$$-\nabla^2\vec{E} = -\frac{\partial}{\partial t}\left(\mu_0\sigma\vec{E} + \mu_0\epsilon_0\frac{\partial\vec{E}}{\partial t}\right)$$

or

$$\boxed{\nabla^2\vec{E} = \mu_0\epsilon_0\frac{\partial^2\vec{E}}{\partial t^2} + \mu_0\sigma\frac{\partial\vec{E}}{\partial t}}$$

Similarly, taking the curl of the last eq'n yields

$$\vec{\nabla}\times(\vec{\nabla}\times\vec{B}) = \vec{\nabla}\times\left(\mu_0\sigma\vec{E} + \mu_0\epsilon_0\frac{\partial\vec{E}}{\partial t}\right) \Rightarrow \vec{\nabla}(\vec{\nabla}\cdot\vec{B}) - \nabla^2\vec{B} = \mu_0\sigma\vec{\nabla}\times\vec{E} + \mu_0\epsilon_0\vec{\nabla}\times\frac{\partial\vec{E}}{\partial t}$$

$\vec{\nabla}\cdot\vec{B} = 0$ and $\vec{\nabla}\times\vec{E} = -\frac{\partial\vec{B}}{\partial t}$, so we get

$$-\nabla^2\vec{B} = \mu_0\sigma\left(-\frac{\partial\vec{B}}{\partial t}\right) + \mu_0\epsilon_0\frac{\partial}{\partial t}\left(-\frac{\partial\vec{B}}{\partial t}\right)$$

so

$$\boxed{\nabla^2\vec{B} = \mu_0\epsilon_0\frac{\partial^2\vec{B}}{\partial t^2} + \mu_0\sigma\frac{\partial\vec{B}}{\partial t}.}$$